Code No: 138FU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
DATA ANALYTICS
(Common to ECE, CSE)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper July/Aug-2023

PART-A

(25 Marks)

1.a)	List the Elements of Data Architecture.	[2]
b)	Which source of Data is more reliable and why?	[3]
c)	Outline the Data Analytics techniques.	[2]
d)	What is the difference between Spark, Scala and Impala?	[3]
e)	What is Model Theory?	[2]
f)	List the BLUE property assumptions.	[3]
g)	What are the methods to measure Forecast accuracy?	[2]
h)	Write about multiple decision tree.	[3]
i)	Define data visualization.	[2]
j)	List out data visualization applications in finance?	[3]

PART-B

(50 Marks)

2.a) List and explain about the Sources of generating Secondary Data.
b) What is an outlier in data set? Illustrate how to detect and remove outliers. [5+5]

OR

3.a) Discuss about Characteristics of effective Data Architecture.
b) Illustrate techniques of missing values treatment. [5+5]

4.a) Explain the application of modeling in business and Elaborate any one.
b) Explain how Key-Value and Column Oriented databases address the volume of the data and the scalability. [5+5]

OR

5.a) Define Data Analytics. Give are the applications of Analytics.
b) Illustrate the Model-Based Techniques for dealing with Missing Data with suitable examples. [5+5]

6. Demonstrate linear regression with suitable example. [10]

OR

7.a) Explain Logistic Regression model and analyze Best Fit Statistics with example.
b) Explain briefly about variable rationalization in Regression modeling. [5+5]

8.a) What is Overfitting? Why Overfitting happens? Give its Preventing techniques?
b) Discuss in detail about STL approach. [5+5]

OR

9.a) Compare and contrast between Supervised and unsupervised Learning.
b) Discuss about Univariate Non-Stationary Processes and its forecast. [5+5]

10.a) Discuss briefly about Icon based Visualization Techniques.
b) Discuss about visualizing the complex data and relations. [5+5]

OR

11. List and explain the steps in Data Visualization in Tableau. [10]

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